MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 23: Cauchy-Riemann equations and Conformal mapping

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

April 5, 2023

Lecture outline

1. Review of Cauchy-Riemann equations
2. Cauchy-Riemann equations in polar coordinates
3. Geometric interpretation of $Arg\, f'(z)$ and $|f'(z)|$.
4. Conformal mapping
5. Fractional-linear transformation

Differentiable function of two variables

We concern mainly with functions defined on some domain $E = G$, and we shall drop the subscript E in $f'_E(z)$ and $d_E f(z)/dz$.
Suppose $u(x, y)$ is a real function of two real variables x and y defined on G.
Then $u(x, y)$ is said to be *differentiable* at the point $(x_0, y_0) \in G$ if $u(x, y)$ can be written in the form

$$\begin{aligned} u(x,y) - u(x_0,y_0) = A(x_0,y_0)(x-x_0) + B(x_0,y_0)(y-y_0) \\ + \varepsilon_1(x,y;x_0,y_0)(x-x_0) + \varepsilon_2(x,y;x_0,y_0)(y-y_0) \end{aligned} \quad (1)$$

where

$$\begin{aligned} \varepsilon_1(x,y;x_0,y_0) &\to 0 \text{ as well as} \\ \varepsilon_2(x,y;x_0,y_0) &\to 0 \text{ as } (x,y) \to (x_0,y_0). \end{aligned}$$

The coefficients $A(x_0, y_0)$ and $B(x_0, y_0)$ in the right-hand side of (1) are just partial derivatives of the function $u(x, y)$ at the point (x_0, y_0):

$$\begin{aligned} A(x_0,y_0) &= \frac{\partial u(x,y)}{\partial x} \Big|_{x=x_0, y=y_0}, \\ B(x_0,y_0) &= \frac{\partial u(x,y)}{\partial y} \Big|_{x=x_0, y=y_0}. \end{aligned}$$

Cauchy-Riemann Theorem. Part 1

Theorem (Cauchy-Riemann equations)

Let

$$f(z) = u(x, y) + iv(x, y)$$

be a function of a complex variable defined on a domain G.

Then a necessary and sufficient condition for $f(z)$ to be differentiable (as a function of a complex variable) at the point $z_0 = x_0 + iy_0 \in G$ is that the functions $u(x, y)$ and $v(x, y)$ be differentiable (as functions of the two real variables x and y) at the point (x_0, y_0) and satisfy the Cauchy-Riemann equations

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \tag{2}$$

at (x_0, y_0).

Cauchy-Riemann Theorem. Part 2

Theorem (Cauchy-Riemann equations)

If conditions 2 are satisfied. $f'(z)$ can be represented in any of the forms:

$$\begin{aligned} f'(z_0) &= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \\ &= \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} \\ &= \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y} \\ &= \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}, \end{aligned} \tag{3}$$

where the partial derivatives are all evaluated at (x_0, y_0).

A function $f(z)$ which is differentiable on a domain G, i.e. at every point of G, is said to be *analytic* (synonymously, holomorphic or regular) on G. If $f(z)$ is analytic in a neighborhood of z_0, $f(z)$ is said to be analytic at z_0.

Proof of Cauchy-Riemann: Necessary Conditions. Part 1

First we prove that the conditions are necessary. Consider the increments

$$\begin{aligned}\Delta z &= z - z_0 \\ &= (x - x_0) + i(y - y_0) \\ &= \Delta x + i\Delta y,\end{aligned}$$

$$\begin{aligned}\Delta f(z) &= f(z) - f(z_0) \\ &= [u(x,y) - u(x_0,y_0)] + i[v(x,y) - v(x_0,y_0)] \\ &= \Delta u + i\Delta v\end{aligned}$$

of the independent variable z and of the function $f(z)$. If $f(z)$ is differentiable at z_0, then

$$\Delta f(z) = f'(z_0)\Delta z + \varepsilon\Delta z, \tag{4}$$

where $\varepsilon \to 0$ as $\Delta z \to 0$.

Proof of Cauchy-Riemann: Necessary Conditions. Part 2

Writing

$$\begin{aligned} f'(z_0) &= a + ib, \\ \varepsilon &= \varepsilon_1 + i\varepsilon_2, \end{aligned}$$

and taking the real and imaginary parts of (4), we find that

$$\begin{aligned} \Delta u &= a\Delta x - b\Delta y + \varepsilon_1 \Delta x - \varepsilon_2 \Delta y, \\ \Delta v &= b\Delta x - a\Delta y + \varepsilon_2 \Delta x - \varepsilon_1 \Delta y, \end{aligned}$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $\Delta x, \Delta y \to 0$, since

$$|\Delta z| = \sqrt{(\Delta x)^2 + (\Delta y)^2}, \quad |\varepsilon_1| \leq |\varepsilon|, \quad |\varepsilon_2| \leq |\varepsilon|.$$

Proof of Cauchy-Riemann: Necessary Conditions. Part 3

It follows that the functions $u(x, y)$ and $v(x, y)$ are differentiable at (x_0, y_0) and

$$\begin{aligned} \frac{\partial u}{\partial x} &= a, \\ \frac{\partial u}{\partial y} &= -b, \\ \frac{\partial v}{\partial x} &= b, \\ \frac{\partial v}{\partial y} &= a. \end{aligned} \tag{5}$$

But (5) immediately implies (2) and (3)

Proof of Cauchy-Riemann: Sufficient Conditions. Part 1

To prove that the conditions of the theorem are sufficient, we reverse the preceding argument. Thus, suppose $u(x, y)$ and $v(x, y)$ are differentiable at the point (x_0, y_0), and suppose (2) holds. Then

$$\begin{aligned} \Delta u &= \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \alpha_1 \Delta x + \alpha_2 \Delta y \\ &= a\Delta x - b\Delta y + \alpha_1 \Delta x + \alpha_2 \Delta y, \end{aligned} \tag{6}$$

$$\begin{aligned} \Delta v &= \frac{\partial v}{\partial x}\Delta x + \frac{\partial v}{\partial y}\Delta y + \beta_1 \Delta x + +\beta_2 \Delta y \\ &= b\Delta x + a\Delta y + \beta_1 \Delta x + \beta_2 \Delta y, \end{aligned} \tag{7}$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2 \to 0$ as $\Delta x, \Delta y \to 0$ and we have written

$$\begin{aligned} a &= \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \\ b &= -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{aligned} \tag{8}$$

Proof of Cauchy-Riemann: Sufficient Conditions. Part 2

Substituting (6) and (7) into the formula $\Delta f(z) = \Delta u + i\Delta v$, we find that

$$\begin{aligned}\Delta f(z) &= a(\Delta x + i\Delta y) + ib(\Delta x + i\Delta y) + (\alpha_1 + i\beta_1)\Delta x + (\alpha_2 + i\beta_2)\Delta y \\ &= (a + ib)\Delta z + \left[(\alpha_1 + i\beta_1)\frac{\Delta x}{\Delta z} + (\alpha_2 + i\beta_2)\frac{\Delta y}{\Delta z}\right]\Delta z\end{aligned} \tag{9}$$

or

$$\Delta f(z) = (a + ib)\Delta z + \varepsilon\Delta z, \tag{10}$$

where

$$\begin{aligned}|\varepsilon| &= \left|(\alpha_1 + i\beta_1)\frac{\Delta x}{\Delta z} + (\alpha_2 + i\beta_2)\frac{\Delta y}{\Delta z}\right| \\ &\leq |(\alpha_1 + i\beta_1)|\left|\frac{\Delta x}{\Delta z}\right| + |(\alpha_2 + i\beta_2)|\left|\frac{\Delta y}{\Delta z}\right| \\ &\leq |(\alpha_1 + i\beta_1)| + |(\alpha_2 + i\beta_2)| \\ &\leq |\alpha_1| + |\beta_1| + |\alpha_2| + |\beta_2|.\end{aligned} \tag{11}$$

Proof of Cauchy-Riemann: Sufficient Conditions. Part 3

Since $\alpha_1, \alpha_2, \beta_1, \beta_2 \to 0$ as $\Delta x, \Delta y \to 0$, it follows that $\varepsilon \to 0$ as $\Delta z \to 0$, and hence, according to (10) $f(z)$ is differentiable at z_0, with derivative

$$f'(z_0) = a + ib. \tag{12}$$

The various representations (3) of $f'(z_0)$ are now immediate consequences of (12) and (8).

Cauchy-Riemann equations in the polar coordinates. Part 1

We will express the differentiability conditions for a function $f(z) = u + iv$ at a point $z \neq 0$ in terms of the polar coordinates

$$\begin{aligned} r &= |z|, \\ \Phi &= Arg\, z. \end{aligned}$$

The appropriate necessary and sufficient conditions for differentiability are that $u(r, \Phi)$ and $v(r, \Phi)$ be differentiable (as functions of two variables r and Φ) and satisfy the polar form of the Cauchy-Riemann equations, i.e.,

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{r}\frac{\partial v}{\partial \Phi}, \\ \frac{\partial v}{\partial r} &= -\frac{1}{r}\frac{\partial u}{\partial \Phi} \end{aligned} \tag{13}$$

at a given nonzero point P.

Cauchy-Riemann equations in the polar coordinates. Part 2

To verify these conditions, we must prove that

1. u and v are differentiable as functions of r and Φ at P if and only if they are differentiable as functions of x and y at P,
2. under these conditions, the equations (13) are equivalent to the equations:

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \tag{14}$$

The first assertion follows from the familiar fact that a differentiable function ($u = u(x, y)$, say) of differentiable functions ($x = r\cos\Phi, y = r\sin\Phi$, say) is also differentiable (with respect to the new variables r and Φ).

Cauchy-Riemann equations in the polar coordinates. Part 3

The second assertion can be verified by direct calculation. For example, if u and v are differentiable functions x and y, and if the equations (14) hold, then

$$\begin{aligned} \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cos\Phi + \frac{\partial u}{\partial y}\sin\Phi &= \frac{\partial v}{\partial y}\cos\Phi - \frac{\partial v}{\partial x}\sin\Phi = \frac{1}{r}\frac{\partial v}{\partial \Phi}, \\ \frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\Phi + \frac{\partial v}{\partial y}\sin\Phi &= -\frac{\partial u}{\partial y}\cos\Phi + \frac{\partial u}{\partial x}\sin\Phi = -\frac{1}{r}\frac{\partial u}{\partial \Phi}. \end{aligned} \tag{15}$$

Writing (15) in the form

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x}\cos\Phi - \frac{\partial v}{\partial x}\sin\Phi, \\ \frac{\partial v}{\partial r} &= \frac{\partial u}{\partial x}\sin\Phi + \frac{\partial v}{\partial x}\cos\Phi, \end{aligned}$$

and solving for $\partial u/\partial x$ and $\partial v/\partial x$, we obtain

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r}\cos\Phi + \frac{\partial v}{\partial r}\sin\Phi, \\ \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial r}\sin\Phi + \frac{\partial v}{\partial r}\cos\Phi, \end{aligned}$$

Cauchy-Riemann equations in the polar coordinates. Part 4

and hence

$$
\begin{aligned}
f'(z) &= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \\
&= \frac{\partial u}{\partial r}(\cos\Phi - i\sin\Phi) + i\frac{\partial v}{\partial r}(\cos\Phi - i\sin\Phi) \\
&= \left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right)(\cos\Phi - i\sin\Phi) \\
&= \frac{r}{z}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right).
\end{aligned}
\tag{16}
$$

This formula is convenient for calculating $f'(z)$ with the help of polar coordinates. Using (13), we can write $f'(z)$ in the form

$$
f'(z) = \frac{1}{z}\left(\frac{\partial v}{\partial \Phi} - i\frac{\partial u}{\partial \Phi}\right). \tag{17}
$$

Geometry of $Arg\, f'(z)$ and $|f'(z)|$. Part 1

Let l be a continuous curve with equation $z = \lambda(t)$, $t \in [a, b]$, and suppose $\lambda(t)$ is differentiable at point $t_0 \in [a, b]$ (relative to the set $[a, b]$). Let $\{t_n\}$ be an arbitrary sequence of points in $[a, b]$ converging to t_0 ($t_n \neq t_0$), and consider the difference quotient

$$r_n = \frac{\lambda(t_n) - \lambda(t_0)}{t_n - t_0}. \tag{18}$$

Obviously $r_n \to r_0 = \lambda'(t_0)$ as $n \to \infty$.

Definition

The curve l is said to have a tangent at the point $z_0 = \lambda(t_0)$ if the limit

$$\theta = \lim_{n\to\infty} Arg\, r_n \tag{19}$$

exists, and then the tangent is said to have inclination θ. Geometrically, the tangent to l at z_0 is represented by the ray τ emanating from z_0 which makes the angle θ with the positive real axis.

Geometry of $Arg\, f'(z)$ and $|f'(z)|$. Part 2

Remarks.

- Formula (19) means that given any $\varepsilon > 0$, there is an integer $N(\varepsilon) > 0$ and a sequence $\{\theta_n\}$, where each θ_n is a value of $Arg\, r_n$, such that $|\theta_n - \theta| < \varepsilon$ for all $n > N(\varepsilon)$. Clearly, θ is only defined to within a multiple of 2π. The angle θ will always be measured from the positive real axis to the tangent τ (in the counterclockwise direction for a positive value of θ).
- Clearly, if $\lambda'(t_0) \neq 0$, l has a tangent at z_0, since

$$\theta = \lim_{n\to\infty} Arg\, r_n = Arg\, r_0 = Arg\, \lambda'(t_0) \tag{20}$$

  and then the inclination of the tangent is just the argument of the complex number $\lambda'(t_0)$. On the other hand, if $\lambda'(t_0) = 0$, l may or may not have been a tangent at z_0, since the fact that $r_0 \to 0$ implies only that $|r_n| \to 0$ and says noting about the behavior of $Arg\, r_n$ However, if l has a tangent at z_0, then $\lambda(t_n) \neq \lambda(t_0)$ for all t_n sufficiently close to t_0, since $Arg\, 0$ is meaningless (This condition is automatically satisfied if $\lambda'(t_0) \neq 0$, since otherwise $r_n = 0$ for t arbitrary close to t_0, which implies that $\lambda'(t_0) = 0$, contrary to hypothesis).

Geometry of $Arg\, f'(z)$ and $|f'(z)|$. Part 3

Remarks.

- As just defined, the tangent is ray, not a vector. If $\lambda'(t_0) \neq 0$, we can also introduce a *tangent vector* to l at z_0, defined as the vector of length $|\lambda'(y_0)|$ which makes the angle θ with the positive real axis.

Theorem

Let G be a domain, and let $f(z)$ be a continuous function of a complex variable defined on G. Suppose $f(z)$ has a nonzero derivative $f'(z_0)$ at a point $z_0 \in G$, and let l be a curve which passes through z_0 and has a tangent τ at z_0. Then $w = f(z)$ maps l into a curve L in the w-plane which passes through the point $w_0 = f(z_0)$ and has a tangent T at w_0. Moreover, the inclination of T exceeds the inclination of τ by the angle $Arg\, f'(z_0)$.

Proof of tangent mapping theorem. Part 1

Suppose l has the equation $z = \lambda(t),\ t \in [a, b]$, and let $z_0 = \lambda(t_0)$. By hypothesis,

$$\theta = \lim_{n\to\infty} Arg\ r_n \tag{21}$$

exists, where r_n is given by (18). The function $w = f(z)$ maps l into a curve L in the w-plane with equation

$$w = f[\lambda(t)] = \Lambda(t), \quad t \in [a, b], \tag{22}$$

where $w_0 = f(z_0) = \Lambda(t_0)$. Let $\{t_n\}$ be an arbitrary sequence of points in $[a, b]$ converging to t_0, and let

$$R_n = \frac{\Lambda(t_n) - \Lambda(t_0)}{t_n - t_0}.$$

Proof of tangent mapping theorem. Part 2

Then the tangent to L at w_0 has inclination

$$\Theta = \lim_{n\to\infty} Arg\, R_n$$

provided this limit exists.

Clearly we have

$$R_n = \frac{\Lambda(t_n) - \Lambda(t_0)}{\lambda(t_n) - \lambda(t_0)} \frac{\lambda(t_n) - \lambda(t_0)}{t_n - t_0},$$

where the first factor in the right-hand side is well-defined, since $\lambda(t_n) \neq \lambda(t_0)$ for all t_n sufficiently close to t_0 (l is assumed to have a tangent at z_0).

Proof of tangent mapping theorem. Part 3

Therefore

$$
\begin{aligned}
\Theta &= \lim_{n\to\infty} Arg\, R_n \\
&= \lim_{n\to\infty} Arg\, \frac{\Lambda(t_n)-\Lambda(t_0)}{\lambda(t_n)-\lambda(t_0)} \frac{\lambda(t_n)-\lambda(t_0)}{t_n-t_0} \\
&= \lim_{n\to\infty} Arg\, \frac{w_n-w_0}{z_n-z_0} r_n \\
&= \lim_{n\to\infty} Arg\, \frac{w_n-w_0}{z_n-z_0} + \lim_{n\to\infty} Arg\, r_n \\
&= Arg \lim_{n\to\infty} \frac{w_n-w_0}{z_n-z_0} + \theta \\
&= Arg \lim_{n\to\infty} f'(z_0) + \theta,
\end{aligned}
\tag{23}
$$

where $w_n = \Lambda(t_n), z_n = \lambda(t_n)$ and θ is the inclination of τ at z_0 (In reversing the order of the operations Arg and $\lim_{n\to\infty}$, we have used the fact that $f'(z_0) \neq 0$.).

Proof of tangent mapping theorem. Part 4

It follows from (23) that Θ exists and that

$$\Theta - \theta = Arg f'(z_0),$$

as asserted. We note that things are particularly simple in the case where $\lambda'(t_0) \neq 0$, since then

$$\begin{aligned} \Theta &= Arg\, \Lambda'(t_0) \\ &= Arg\, [f'(z_0)\lambda'(t_0)] \\ &= Arg\, [f'(z_0)] + Arg\, [\lambda'(t_0)] \\ &= Arg\, [f'(z_0)] + \theta, \end{aligned} \tag{24}$$

by the rule for differentiating the composite function (22).

Conformal mapping. Part 1

Now let l_1 and l_2 be two curves with a common initial point z_0, which have tangents τ_1 and τ_2 at z_0 and suppose the angle between τ_1 and τ_2 is measured from τ_1 to τ_2. Suppose l_1 and l_2 have images L_1 and L_2 under $f(z)$. Then, according to Tangent Mapping Theorem (4), if $f'(z_0) \neq 0$, L_1 and L_2 have tangents T_1 and T_2 at the point $w_0 = f(z_0)$, where T_1 and T_2 are obtained by rotating τ_1 and τ_2 through the same angle $Arg\, f'(z_0)$. Therefore the angle between L_1 and L_2 equals the angle between l_1 and l_2, and is measured in the same direction, i.e. from L_1 to L_2. In other words, a continuous function $w = f(z)$ with a nonzero derivative $f'(z_0)$ maps all curves in the z-plane which pass through z_0, and have tangents at z_0 into curves in the w-plane which pass through $w_0 = f(z_0)$ and have tangents at w_0, and moreover, *the mapping preserves angles between curves*. A mapping by a continuous function which preserves angles between curves passing through a given point z_0 is said to be *conformal* at z_0. If a conformal mapping preserves the directions in which angles are measured (as well as their magnitudes), it is called a *conformal mapping of the first kind*, but if it reverses the directions in which angles are measured, it is called a *conformal mapping of the second kind*.

Conformal mapping. Part 2

Thus Tangent Mapping Theorem (4) has the following consequence:

Theorem

Let G be a domain, and let $f(z)$ be an analytic function on G. Then $f(z)$ is a conformal mapping of the first kind at every point of G where $f'(z) \neq 0$.

As we have seen, $Arg\, f'(z_0)$ represents the rotation undergone by the tangent to a curve l at the point $z_0 \in l$ when transforming to the new curve $L = f(l)$ and the new point $w_0 = f(z_0)$ and the new point $w_0 = f(z_0)$. In particular, if $f'(z_0)$ is a positive real number, the tangents to l at z_0 and to L at w_0 are parallel and point in the same direction.
To explain the geometric meaning of the quantity $|f'(z_0)|$, i.e., the absolute value of the derivative at z_0, we note that

$$|f'(z_0)| = \lim_{z \to z_0} \frac{|f(z) - f(z_0)|}{|z - z_0|}.$$

Conformal mapping. Part 3

The numbers $|z - z_0|$ and $|f(z) - f(z_0)|$ are the distance between the points z and z_0 in the z-plane, and the distance between their images $f(z)$ and $f(z_0)$ in the w-plane, respectively. Thus, interpreting

$$\frac{|f(z) - f(z_0)|}{|z - z_0|}$$

as the *linear magnification ratio* (or simply the *magnification*) of the vector $z - z_0$ under the mapping $w = f(z)$, we can regard $|f'(z_0)|$ as the *magnification at the point* z_0 under $w = f(z)$ (Here the word *magnification* is used in general sense, and can correspond to *stretching* if $|f'(z_0)| > 1$ or *shrinking* if $|f'(z_0)| < 1$, or neither if $|f'(z_0)| = 1$).

Remark.
The size of magnification at the point z_0 does not depend on the choice of the finite vector $z - z_0$ drawn from z_0 since $|f'(z_0)|$ is not the actual magnification of any such vector, but rather a limiting magnification as $z \to z_0$.

Example: Fractional-linear transformation. Part 1

We now examine the *fractional linear transformation* or *Möbius transformation*

$$L(z) = \frac{az + b}{cz + d}, \tag{25}$$

where a, b, c, d are arbitrary complex numbers (except that c and d are not both zero). First suppose that $c = 0$. Then $L(z)$ reduces to

$$L(z) = \alpha z + \beta \ \ (\alpha = a/d, \beta = b/d) \tag{26}$$

and it sometimes called the *entire linear transformation*. The transformation (26) is defined for all values of z, and if $\alpha \neq 0$, its derivative $L'(z)$ is a nonzero constant, so that (26) is conformal at every point of the (finite) z-plane. Under this transformation, the tangents to all curves in the z-plane are rotated through the same angle, equal to $Arg\, \alpha$, and the magnification at every point equals $|\alpha|$.

Fractional-linear transformation. Part 2

If $\alpha = 1$, then

$$Arg\,\alpha = 2k\pi, \quad |\alpha| = 1, \tag{27}$$

where k is an integer, and then both the rotation and expansion produce no effect. In this case, the transformation takes the form

$$w = z + \beta, \tag{28}$$

which obviously corresponds to displacing the whole plane by the vector β. On the other hand, if $\alpha \neq 1$ (and $\alpha \neq 0$), the transformation (26) can be written in the form

$$w - z_0 = \alpha(z - z_0),$$

where z_0 is determined from the equation

$$z_0 = \alpha z_0 + \beta$$

Fractional-linear transformation. Part 3

Obviously, z_0 is invariant under the transformation (26), i.e., z_0 is a *fixed point* of the transformation. If $\alpha \neq 0$, the point at infinity is also a fixed point.

Then it is immediately clear that the transformation (26) is equivalent to a rotation of the whole plane through the angle *Arg* α about the point $z_0 = \beta/(1-\alpha)$, together with a uniform magnification by the factor $|\alpha|$ relative to the point z_0. This magnification is sometimes called a *homothetic transformation* (or *transformation of similitude*) with *ray center* z_0 and *ray ratio* $|\alpha|$.

Next suppose that $c \neq 0$ in (25). Then the derivative

$$L'(z) = \frac{ad - bc}{(cz + d)^2} = \frac{ad - bc}{c^2} \frac{1}{(z - \delta)^2}$$

exists, if $z \neq \delta$, where $\delta = -d/c$.

Fractional-linear transformation. Part 4

If the determinant

$$ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{29}$$

vanishes, then the rows are proportional, i.e.,

$$\frac{a}{c} = \frac{b}{d} = \mu$$

or

$$a = \mu c, \quad b = \mu d,$$

where μ is a constant, so that (25) reduces to the trivial transformation

$$L(z) = \frac{az + b}{cz + d} = \frac{\mu cz + \mu d}{cz + d} = \mu$$

Fractional-linear transformation. Part 5

If $ad - dc \neq 0$, then $L'(z) \neq 0$ for all $z \neq \delta$, and hence the mapping $w = L(z)$ is conformal at all finite points except possibly at $z = \delta$. Under the mapping, the tangents to curves passing through any point $z \neq \delta$ are rotated through an angle equal to

$$Arg\, L'(z) = Arg\, \frac{ad - bc}{c^2} - 2Arg(z - \delta),$$

while the magnification at z equals

$$|L'(z)| = \left| \frac{ad - bc}{c^2} \right| \frac{1}{|z - \delta|^2}.$$

Fractional-linear transformation. Part 6

The angle through which tangents are rotated has the same value for all points with equal values of $Arg\,(z-\delta)$, i.e., along any ray drawn from δ, but otherwise varies from point to point. Similarly, in general the magnification varies with z, but it has the same value for all points with equal values of $|z-\delta|$, i.e., along any circle with center δ. In particular, the magnification is equal to 1 at every point of the circle C with equation

$$|z-\delta| = \frac{1}{|c|}\sqrt{|ad-bc|}$$

(called the *isometric circle* of the Möbius transformation), is greater than 1 inside C (approaching ∞ as $z \to \delta$), and is less than 1 outside C (approaching 0 as $z \to \infty$).

Conformal mapping of the extended plane. Part 1

As in the preceding example, let $c \neq 0$ and $ad - bc \neq 0$. Then it is clear that

$$\begin{aligned} \lim_{z \to \delta} \frac{az+b}{cz+d} &= \infty, \\ \lim_{z \to \infty} \frac{az+b}{cz+d} &= \frac{a}{c} = A, \end{aligned}$$

where $\delta = -d/c$. Suppose we complete the definition of $L(z)$ by setting

$$L(\delta) = \infty, \quad L(\infty) = A. \tag{30}$$

Then $L(z)$ is defined in the while extended plane, and maps the finite point δ into ∞ (the point at infinity) and ∞ into the point A. We now show that the mapping $w = L(z)$ is conformal at the points δ and A (it has already been shown that the mapping is conformal everywhere else). First we must suitably define the concept of an angle with its vertex at infinity.

Conformal mapping of the extended plane. Part 2

Definition

Two continuous curves γ_1 and γ_2 in the extended plane form an angle of α radians with its vertex at infinity if and only if their images $\widetilde{\gamma_1}$ and $\widetilde{\gamma_2}$ in the extended plane under the transformation $\zeta = 1/z$ form an angle of α radians with its vertex at the origin.

Returning to the mapping $w = L(z)$, let γ_1 and γ_2 be two curves forming an angle θ with its vertex at the point δ, and let Γ_1 and Γ_2 be their images in the w-plane. To prove that Γ_1 and Γ_2 form an angle θ with its vertex at infinity, we subject the w-plane to the transformation

$$\eta = \frac{1}{w}.$$

Conformal mapping of the extended plane. Part 3

The the curves Γ_1 and Γ_2 go into two curves Γ_1^* and Γ_2^*, and the point at infinity goes into the origin of coordinates. Obviously, we can go from γ_1 and γ_2 in the z-plane to Γ_1^* and Γ_2^* in the η-plane by making the Möbius transformation

$$\eta = \frac{cz + d}{az + b},$$

which is conformal at the point $z = \delta = -d/c$. It follows that the curves Γ_1^* and Γ_2^* form an angle θ with its vertex at the origin. Therefore the curves Γ_1 and Γ_2 also form an angle θ with its vertex at infinity. This proves that the mapping $w = L(z)$ is conformal at the point $z = \delta$.

Conformal mapping of the extended plane. Part 4

The fact that $w + L(z)$ is conformal at ∞ is proved similarly. In fact, if the curves γ_1 and γ_2 go through the point at infinity in the z-plane, their images Γ_1 and Γ_2 in the w-plane go though the point A. Suppose γ_1 and γ_2 form an angle θ with its vertex at infinity. This means that their images γ_1^* and γ_2^* under the transformation $\zeta = 1/z$ form an angle θ with its vertex at the origin. But we can obviously go from γ_1^* and γ_2^* to Γ_1 and Γ_2 by making the Möbius transformation

$$\begin{aligned} w &= \frac{az+b}{cz+d} \\ &= \frac{a\frac{1}{\zeta}+b}{c\frac{1}{\zeta}+d} \\ &= \frac{a+b\zeta}{c+d\zeta}, \end{aligned} \tag{31}$$

which is conformal at the point $\zeta = 0$. It follows that Γ_1 and Γ_2 form the same angle θ at the point $A = a/c$. This proves that the mapping $w = L(z)$ is conformal at ∞. The situation can be summarized by saying that *the transformation $w + L(z)$ is a conformal mapping of the extended plane onto itself.*

Conformal mapping of the extended plane. Part 5

These considerations suggest the following definition:

Definition

A function $f(z)$ is said to be *analytic* at $z = \infty$ if the function $f^*(\zeta) = f(1/\zeta)$ is analytic at $\zeta = 0$. In particular, if $f(z)$ is analytic at $z = \infty$, the limit

$$\lim_{z\to\infty} f(z) = \lim_{\zeta\to 0} f^*(\zeta) = f(\infty)$$

always exists and is finite. We define the derivative of $f(z)$ at $z = \infty$ to be the quantity

$$f'(\infty) = f^{*\prime}(0),$$

where it should be noted that in general

$$f'(\infty) \neq lim_{z\to\infty} f'(z).$$

Conformal mapping of the extended plane. Part 6

Then, by the argument above for the special case of the Möbius transformation, it is easy to see that the mapping $w = f(z)$ is conformal at ∞ if $f'(\infty) \neq 0$. With this approach, the conformality at ∞ of the Möbius transformation

$$L(z) = \frac{az + b}{cz + d} \quad (ad - bc \neq 0, c \neq 0)$$

follows from the fact that

$$L'(\infty) = -\frac{ad + bc}{c^2} \neq 0.$$

Remark. Similarly, if

$$\lim_{z \to a} f(z) = \infty,$$

but if

$$\varphi(z) = \frac{1}{f(z)}$$

is analytic at $z = a$, with derivative $\varphi'(a) \neq 0$, then, just as in the case of the Möbius transformation, the mapping $w = f(z)$ is conformal at $z = a$.

Thank you